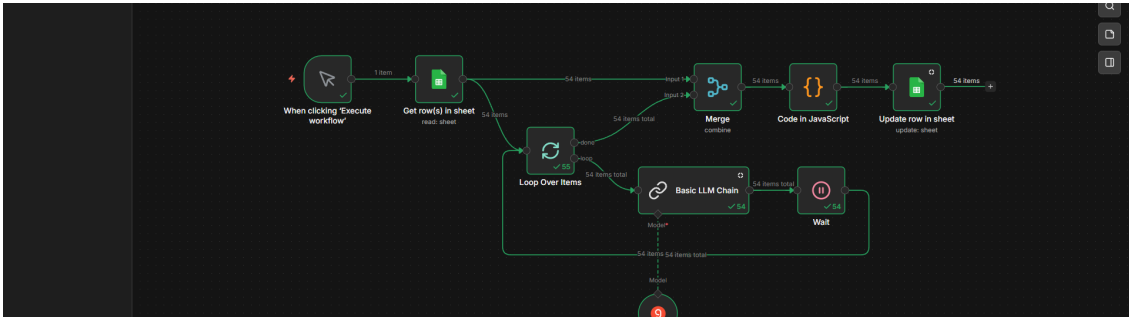




AI Customer Intelligence Prototype - Utkarsh Shirbhate

[Video Link](#)

1. Project Title & Overview

Item	Detail
Title	Beastlife AI Customer Intelligence Prototype
Objective	To automate the categorization of customer support queries across multiple platforms (Instagram, WhatsApp, Email) to identify operational bottlenecks.

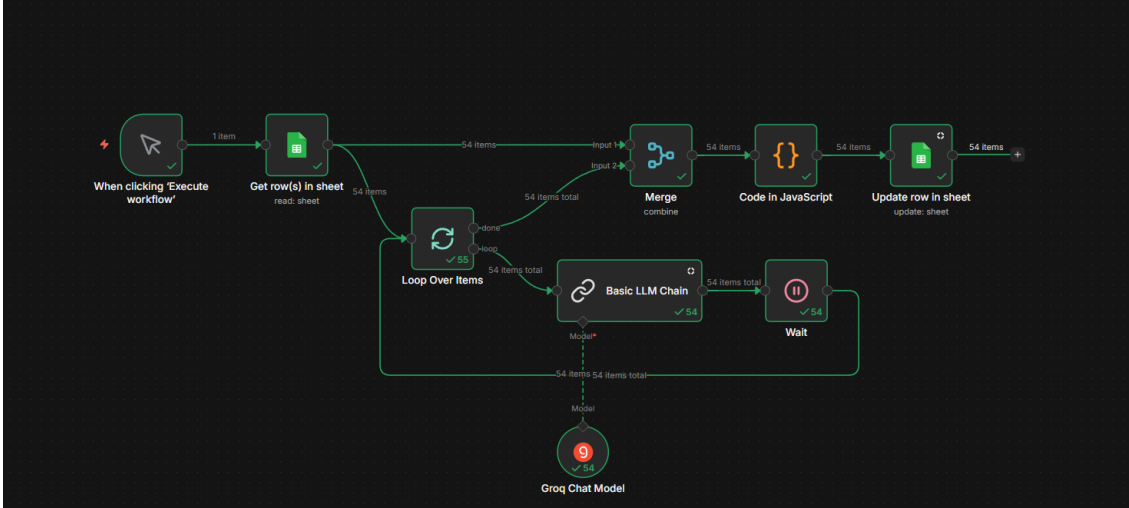
2. Workflow Architecture

The Customer Intelligence workflow is built to systematically process raw customer support tickets, categorize them using a large language model (LLM), and store the structured data for analysis.

Component	Tool/Model	Function
Automation	n8n	Orchestrates the entire data pipeline.

Component	Tool/Model	Function
AI Intelligence	Groq	Performs fast, cost-efficient query categorization and confidence scoring.
Database	Google Sheets	Stores raw and processed (categorized) customer data.
Visualization	Looker Studio	Visualizes category distribution and operational bottlenecks.

Architecture Explanation: The Logic Built

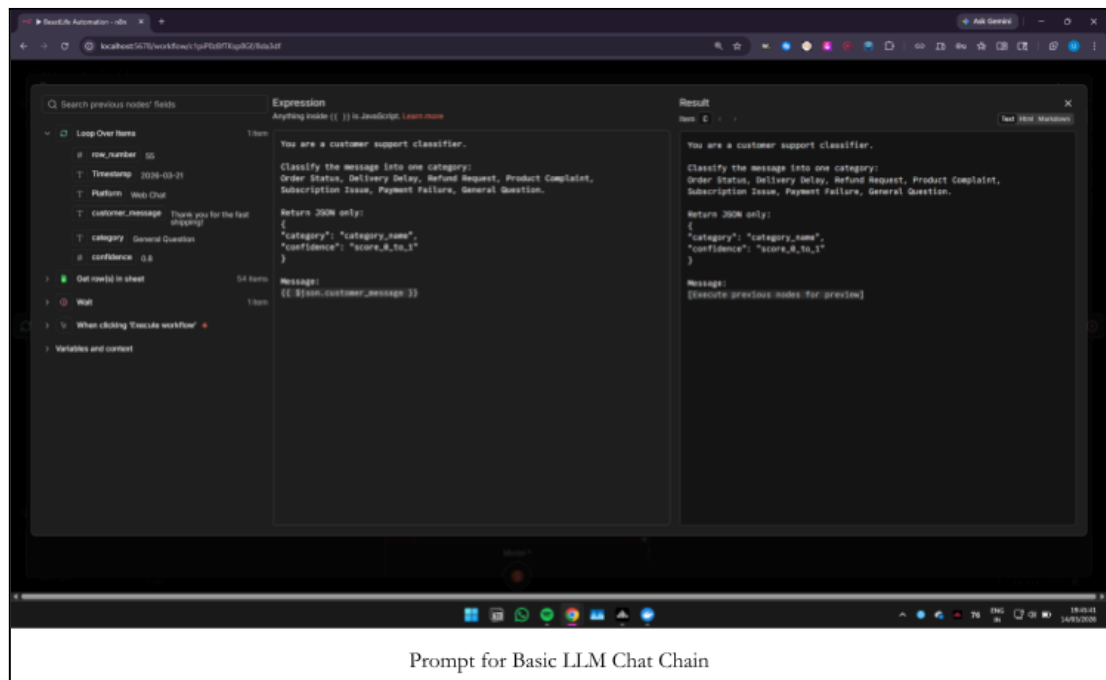


The workflow logic is designed around efficiency and adherence to API best practices:

1. **The Trigger:** The workflow starts by fetching 50+ rows of raw customer data from a source (Google Sheets or a simulated API).
2. **The Intelligence:** A specialized node sends the raw query text to the Groq API. It utilizes a fine-tuned system prompt instructing the LLM (Llama-3.1-8b) to extract both the appropriate support **category** and a **confidence score** for the categorization, ensuring the output is consistently formatted as a JSON object.

3. **Data Storage:** The output (Raw Query, Category, Confidence) is written back to the Google Sheet, creating a structured dataset for the dashboard.

Google Sheets → n8n Workflow → Groq LLM Classification → Structured Data Storage → Looker Studio Dashboard



LLM Classification Prompt

The Groq LLaMA-3.1 model receives the raw customer query and classifies it into one of seven predefined support categories.

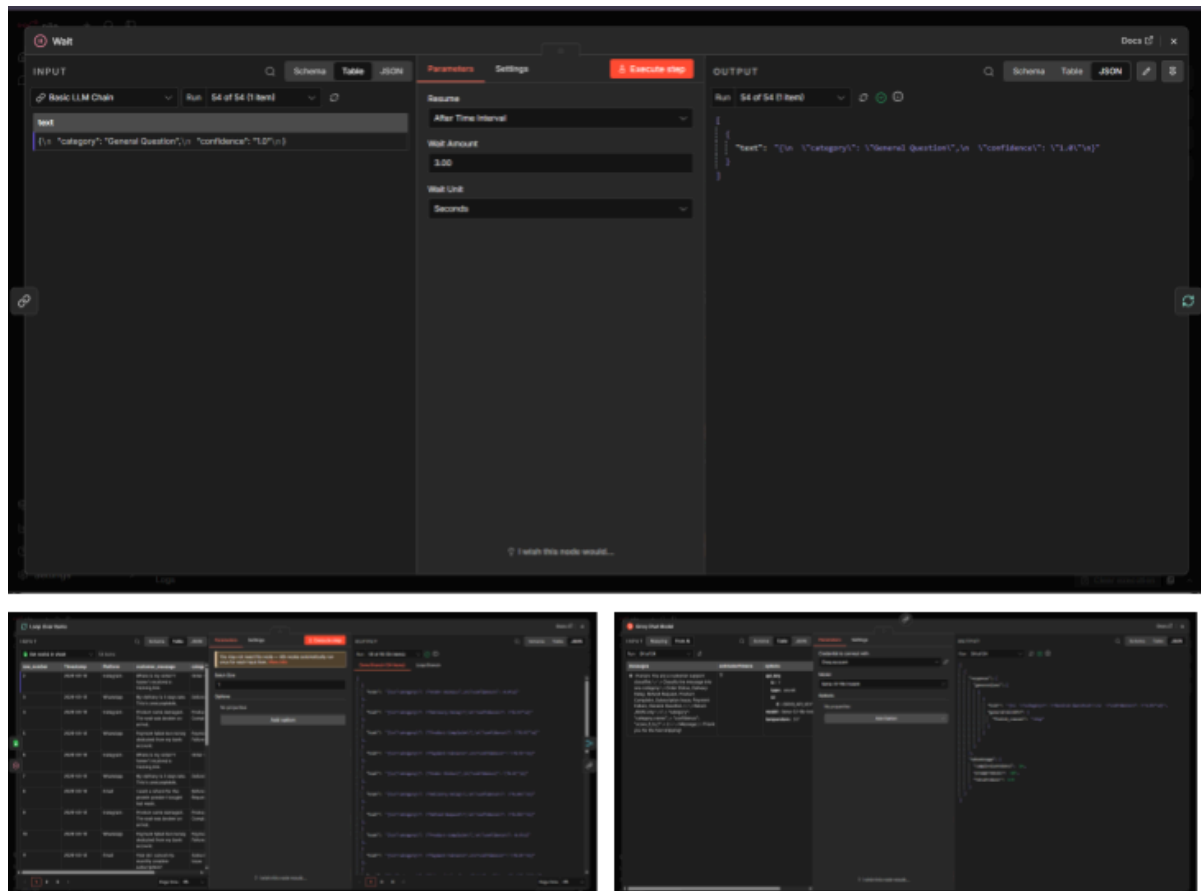
Example output:

```
{  
  "category": "Order Status",  
  "confidence": 0.91  
}
```

Scaling & Rate Limiting: Batch Throttling Implementation

To ensure system stability and respect the Groq API limit of 30 requests per minute (RPM), a robust scaling mechanism was implemented. This involves

Batch Throttling within the n8n canvas. The workflow processes data in small batches, with a deliberate delay of **1 item per 3 seconds**. This rate (20 items per minute) provides a comfortable buffer below the 30 RPM limit, preventing HTTP 429 "Too Many Requests" errors and ensuring the system remains scalable for processing thousands of rows without failure.



Rate limiting for Groq API

3. Automation Opportunities

Analysis of the dashboard results shows that "Product Complaints" (22.2%) and "Delivery Delays" (16.7%) are the two highest categories, indicating significant operational bottlenecks. The AI categorisation prototype immediately points to high-impact automation solutions:

Category	High-Impact Solution	Technical Implementation
Order Status & Delivery Delay	Solution 1 (Auto-Replies): Reduce manual tickets by providing instant tracking.	The AI can be linked to a logistics API (like ShipStation) to automatically fetch and provide real-time tracking links. This is expected to reduce manual tickets by ~35%.
General Questions	Solution 2 (Instant FAQ): Improve first-response resolution time.	An AI Agent can be implemented using RAG (Retrieval-Augmented Generation) technology, utilizing the Beastlife website data as its knowledge base to answer product-specific questions instantly.
Refund Requests	Solution 3 (Smart Escalation): Accelerate the approval process for high-value or high-urgency customers.	These tickets can be automatically routed to a senior human agent. The AI would generate a summary of the customer's previous interactions and the request, providing the agent with full context upfront.

4. Scaling Strategy

To transition this prototype into a production-grade, enterprise system capable of handling high query volumes and maintaining high reliability, the following strategies are required:

Component	Production Upgrade	Rationale
Volume Handling	Transition from Google Sheets to PostgreSQL or BigQuery.	A high-performance database is necessary to handle 10,000+ queries per

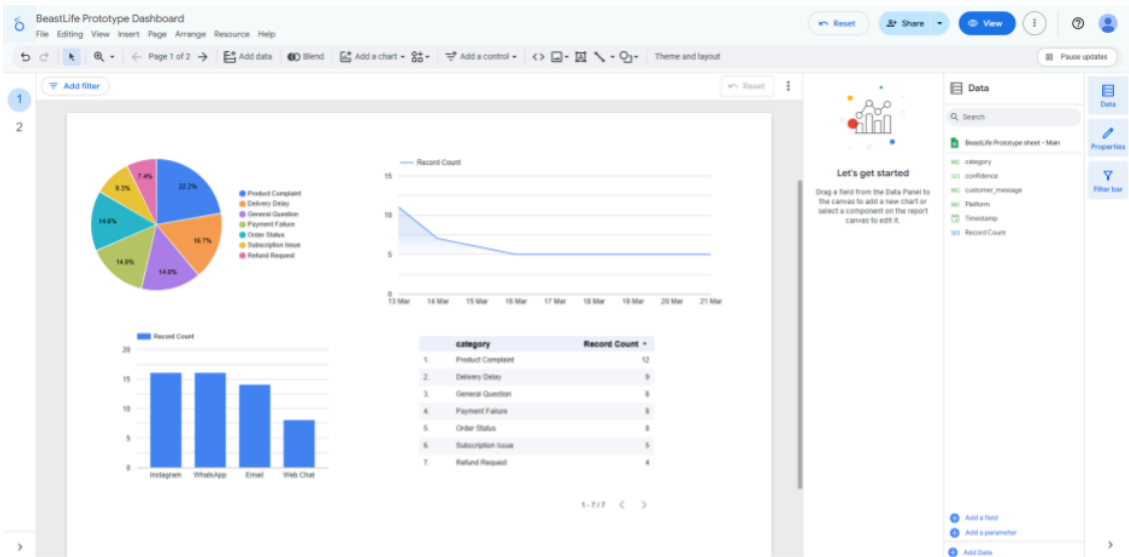
Component	Production Upgrade	Rationale
		day/week, ensuring rapid read/write access and data integrity under load.
Reliability	Implement an "Error Fallback" LLM model.	Implementing a fallback (e.g., if Groq fails, the system automatically routes the request to OpenAI) ensures 100% uptime for customer support intelligence, maintaining business continuity.

5. Final Deliverables List

The successful implementation of the Beastlife AI Customer Intelligence Prototype is verified by the following deliverables:

The Dashboard

[A bar chart representing the distribution of customer support query categories, with 'Product Complaints' at 22.2% and 'Delivery Delays' at 16.7%]

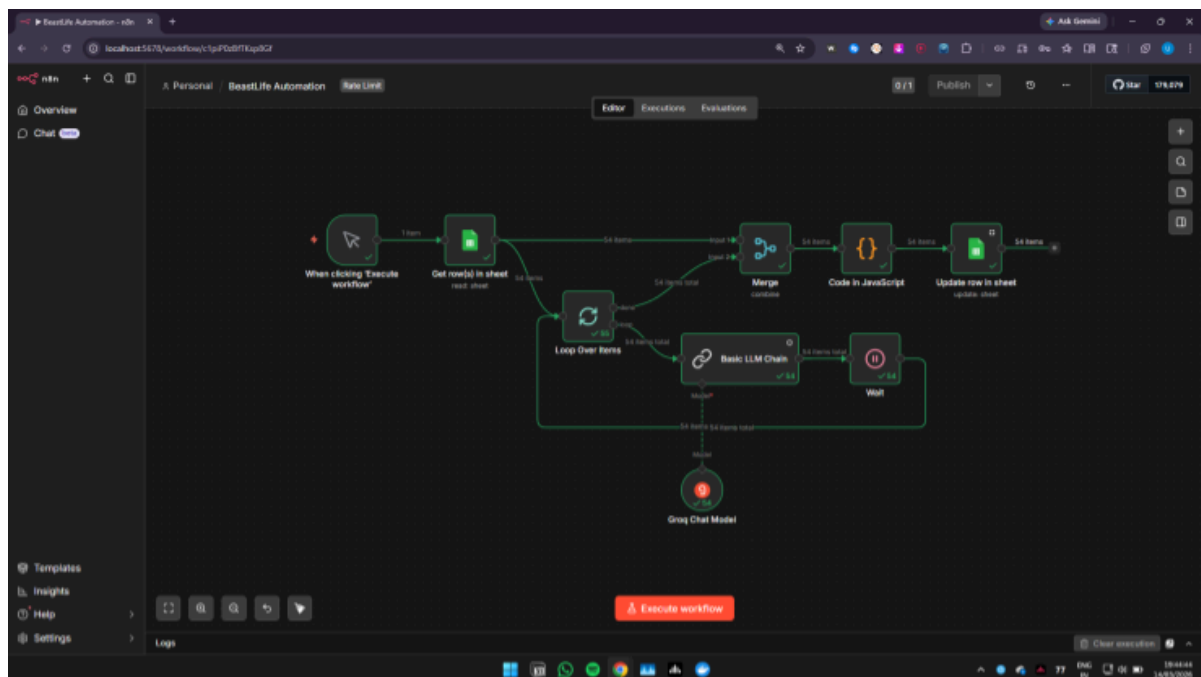


Looker Studio Dashboard

Key Insights

1. Product complaints represent the largest category (22.2%), suggesting product quality or expectation mismatch.
2. Delivery delays account for 16.7% of tickets, indicating logistics communication gaps.
3. Instagram and WhatsApp generate the majority of queries, suggesting automation should prioritize these channels.

The Workflow



Workflow

The Sample Data

	A	B	C	D	E
1	2024-03-03	Instagram	Where is my order? I haven't received a tracking link.	Order Status	0.9
2	2024-03-03	WhatsApp	My delivery is 3 days late. This is unacceptable.	Delivery Delay	0.9
3	2024-03-03	Instagram	Product came damaged. The seal was broken on arrival.	Product Complaint	0.8
4	2024-03-03	WhatsApp	Payment failed but money deducted from my bank account.	Payment Failure	0.9
5	2024-03-03	Instagram	Where is my order? I haven't received a tracking link.	Order Status	0.8
6	2024-03-03	WhatsApp	My delivery is 3 days late. This is unacceptable.	Delivery Delay	0.85
7	2024-03-03	Email	I want a refund for the protein powder I bought last week.	Refund Request	0.8
8	2024-03-03	Instagram	Product came damaged. The seal was broken on arrival.	Product Complaint	0.9
9	2024-03-03	WhatsApp	Payment failed but money deducted from my bank account.	Payment Failure	0.9
10	2024-03-03	Email	How do I cancel my monthly creative subscription?	Subscription Issue	0.9
11	2024-03-03	Web Chat	Are your products vegan friendly?	General Question	1
12	2024-03-14	Instagram	Tracking says delivered but I don't see anything at my door.	Delivery Delay	0.8
13	2024-03-14	WhatsApp	Can I change my shipping address for order #10567?	Order Status	0.8
14	2024-03-14	Email	I was charged twice for the same transaction. Help!	Payment Failure	1
15	2024-03-14	Instagram	The shaker bottle I ordered is leaking from the lid.	Product Complaint	0.9
16	2024-03-14	WhatsApp	Still waiting for my refund. It's been 10 days.	Refund Request	0.9
17	2024-03-14	Web Chat	Do you ship to remote areas in Hinchinbrook?	General Question	0.9
18	2024-03-14	Email	I want to upgrade my subscription to the Pro plan.	Subscription Issue	0.9
19	2024-03-15	Instagram	Wrong item received. I ordered Whey but got Isolate.	Product Complaint	0.9
20	2024-03-15	WhatsApp	My coupon code BEAST123 isn't working at checkout.	Payment Failure	0.8
21	2024-03-15	Email	Order #1012 has been stuck in 'Processing' for a week.	Delivery Delay	0.8
22	2024-03-15	Instagram	It's ready to my DM regarding the missing 1-shed.	Product Complaint	0.8
23	2024-03-15	WhatsApp	The app crashed during payment. Did it go through?	Payment Failure	0.8
24	2024-03-15	Web Chat	What is the expiry date on the current batch of pre-workout?	Product Complaint	0.8
25	2024-03-16	Email	I need an invoice for my last three orders.	Order Status	0.8
26	2024-03-16	Instagram	Your pre-workout gives me jitters. Can I return it?	Order Status	0.8
27	2024-03-16	WhatsApp	Delivery guy was rude and didn't deliver the package.	Delivery Delay	0.8
28	2024-03-16	Email	My subscription renewed but I wanted to pause it.	Subscription Issue	0.9
29	2024-03-16	Web Chat	Is it safe to take this with milk or only water?	Product Complaint	0.8
30	2024-03-17	Instagram	Box was totally crushed. Contents might be ruined.	Product Complaint	0.9
31	2024-03-17	WhatsApp	Checking status for my exchange request.	Order Status	0.85
32	2024-03-17	Email	I can't decide! Multiple items despite having Isolate.	Payment Failure	0.8
33	2024-03-17	Instagram	Love the product! When is the chocolate flavor back?	Product Complaint	0.5
34	2024-03-17	WhatsApp	Package is in my city but delivery date keeps changing.	Delivery Delay	0.8
35	2024-03-18	Email	I want to delete my account and all my data.	Subscription Issue	0.9
36	2024-03-18	Web Chat	Can I get a bulk discount for my gym?	General Question	0.9
37	2024-03-18	Instagram	My order was supposed to be 2kg delivery. It's been 48h.	Delivery Delay	1
38	2024-03-18	WhatsApp	Money was cut, order not confirmed. Frustrating.	Payment Failure	0.8
39	2024-03-18	Email	This shaker I ordered was defective and won't work.	Product Complaint	0.8

Before running the Workflow

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After running the Workflow

Workflow Demonstration Video

A short screen recording demonstrating the workflow execution and the dataset transformation can be viewed here:

[Video LINK](#)